PRODUCT OPTIMIZATION & REVENUE CONTRIBUTION ANALYSIS FOR AFFICIONADO COFFEE ROASTERS

Corresponding Author : Pratham Rangoonwala

Data Analytics Project (2026)

Abstract

This study analyzes transaction-level sales data from Afficionado Coffee Roasters to evaluate product performance, revenue contribution, and menu efficiency. Using data analytics techniques such as aggregation, Pareto analysis, and visualization, the study identifies key revenue-driving products, category dependencies, and underperforming items. The findings reveal that a small subset of products contributes disproportionately to total revenue, indicating a high level of revenue concentration. The analysis provides actionable recommendations for menu optimization, pricing strategy, and operational efficiency.

1. Introduction

In the specialty coffee retail industry, product performance varies significantly across categories and variants. Businesses often face challenges such as:

- High-selling products generating low revenue
- Premium products with low demand
- Overcrowded menus that reduce operational efficiency

Afficionado Coffee Roasters aims to improve its product strategy by leveraging data-driven insights. This study focuses on understanding **what customers buy and what drives revenue**, enabling better decision-making in menu design and merchandising.

2. Dataset Description

The dataset consists of transaction-level records for the year 2025.

Key Features:

Column Name	Description
transaction_id	Unique transaction identifier
transaction_time	Time of purchase
transaction_qty	Quantity sold
unit_price	Price per unit
product_category	Category (Coffee, Tea, Chocolate)
product_type	Sub-category

product_detail	Product variant
store_location	Store location

Data Preparation Steps:

- Removed invalid or missing timestamps
- Converted time into hourly format
- Computed revenue using:
  Revenue = Quantity × Unit Price
- Filtered inconsistent or missing values

3. Methodology

The analysis follows a structured approach:

Revenue Computation

- Calculated revenue at transaction level
- Aggregated revenue across:
  - Products
  - Product types
  - Categories

Product Popularity Analysis

- Measured total units sold per product
- Ranked products based on sales volume

Revenue Contribution Analysis

- Calculated revenue share (%) of each product
- Compared sales rank vs revenue rank

Pareto Analysis (80/20 Rule)

- Identified products contributing to 80% of revenue
- Measured revenue concentration

Product Efficiency

- Evaluated performance using:

Efficiency Score = Total Revenue / Total Quantity

Visualization Techniques

- Bar charts (Top & Bottom products)
- Scatter plots (Sales vs Revenue)
- Heatmaps (Demand patterns)
- Treemap (Category contribution)

4. Key Performance Indicators (KPIs)

- Product Revenue Contribution (%)
- Product Sales Volume
- Category Revenue Share (%)
- Revenue Concentration Ratio (%)
- Product Efficiency Score

5. Analysis & Findings

5.1 Product Performance

- Top-performing products generate a significant share of total revenue
- Low-performing products contribute minimally despite occupying menu space

- **Insight:**
  A small number of products act as **revenue drivers ("Hero Products")**

5.2 Sales vs Revenue Analysis

- Some products show high sales volume but low revenue
- Others generate high revenue with fewer transactions

- **Insight:**
  Pricing strategy plays a crucial role in profitability

5.3 Category Contribution

- Certain categories dominate revenue contribution
- Dependence on specific categories indicates business risk

- **Insight:**
  Over-reliance on one category may impact long-term stability

5.4 Revenue Concentration (Pareto Analysis)

- A limited number of products contribute ~80% of revenue
- Majority of products fall into the long-tail category

- **Insight:**
  Menu is highly concentrated → opportunity for optimization

5.5 Product Segmentation

Products are classified into:

- **Hero Products:** High sales + high revenue
- **Potential Products:** Moderate performance
- **Low Performers:** Low sales + low revenue

- **Insight:**
  Not all products deserve equal focus

5.6 Demand Patterns

- Revenue varies significantly across hours
- Peak hours show high customer activity

- **Insight:**
  Time-based promotions can improve sales

6. Business Insights

- Revenue is driven by a small set of high-performing products
- Many products contribute very little to total revenue
- Pricing inefficiencies exist in high-volume products
- Demand patterns vary across time and categories

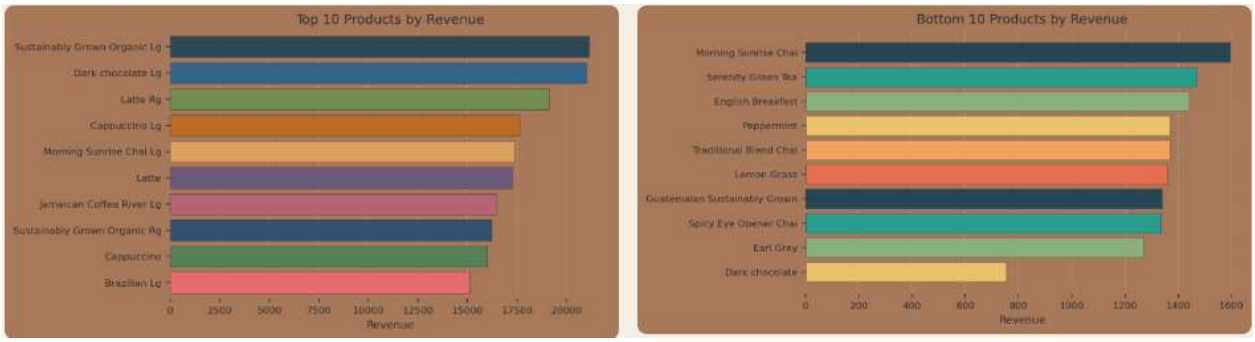

Figure 1: Top 10 and bottom 10 products based on total revenue

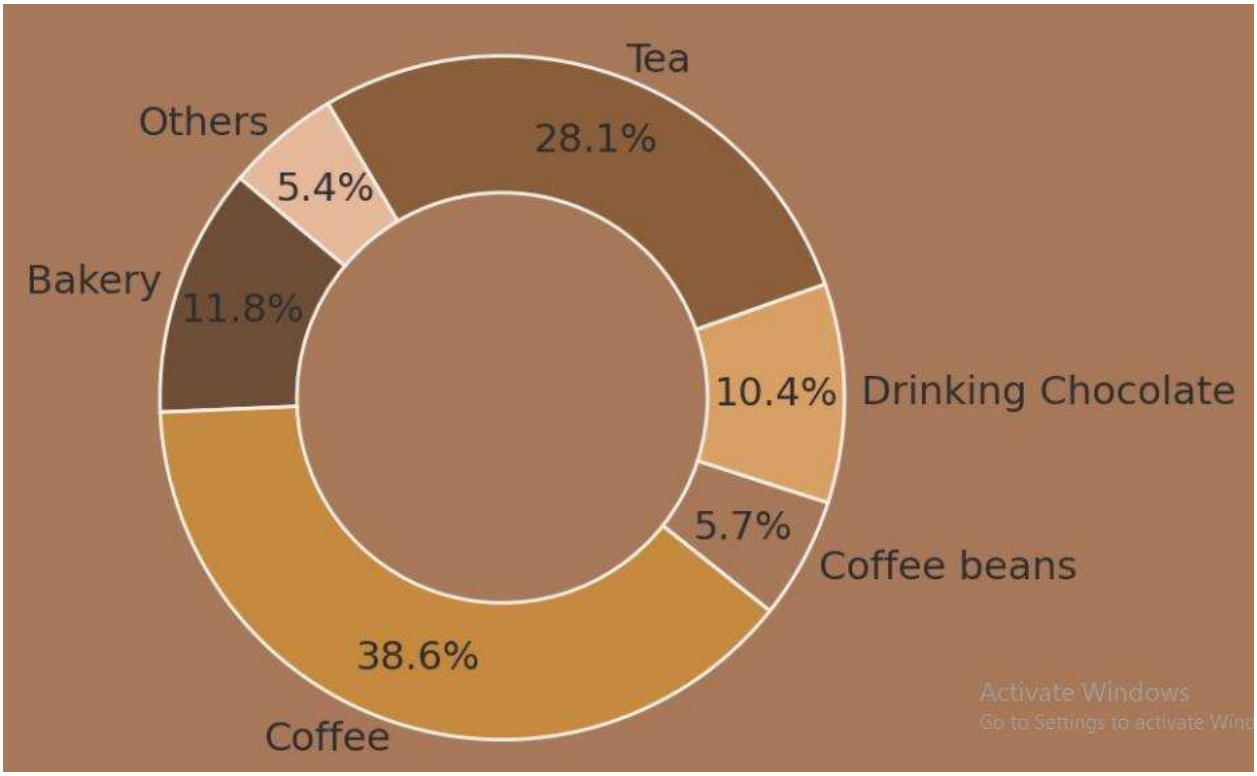

Figure 2: Revenue share across product categories

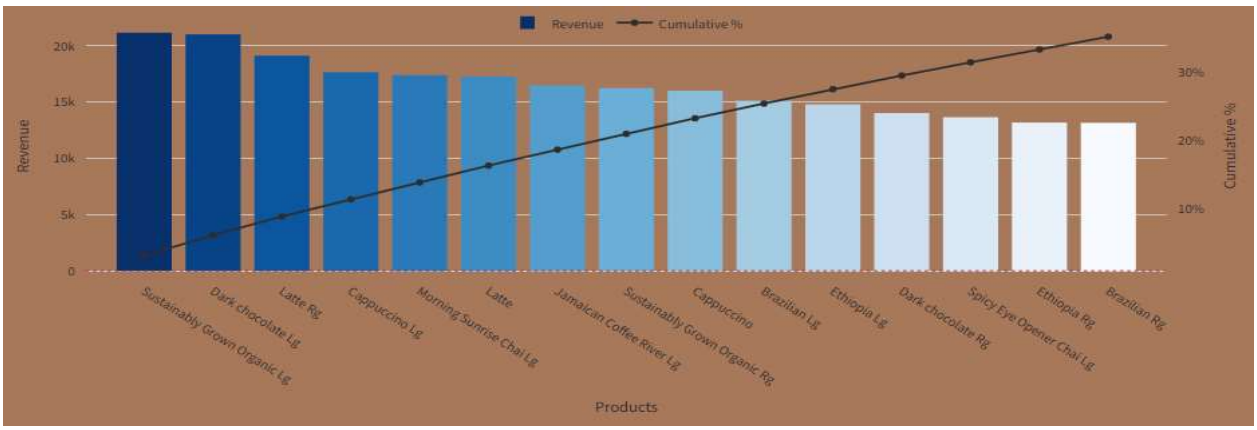

Figure 3: Cumulative revenue contribution of products (80/20 rule)

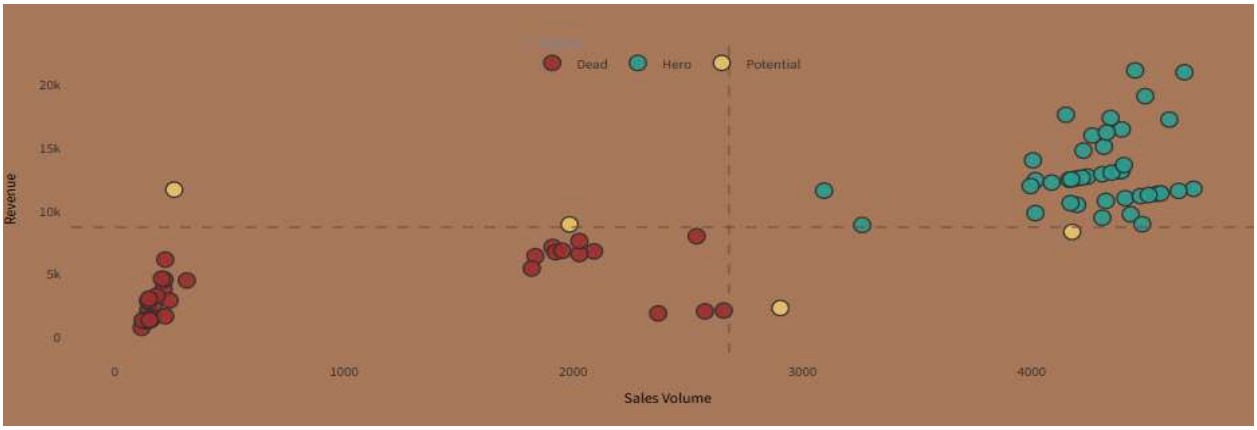

Figure 4: Relationship between product popularity and revenue

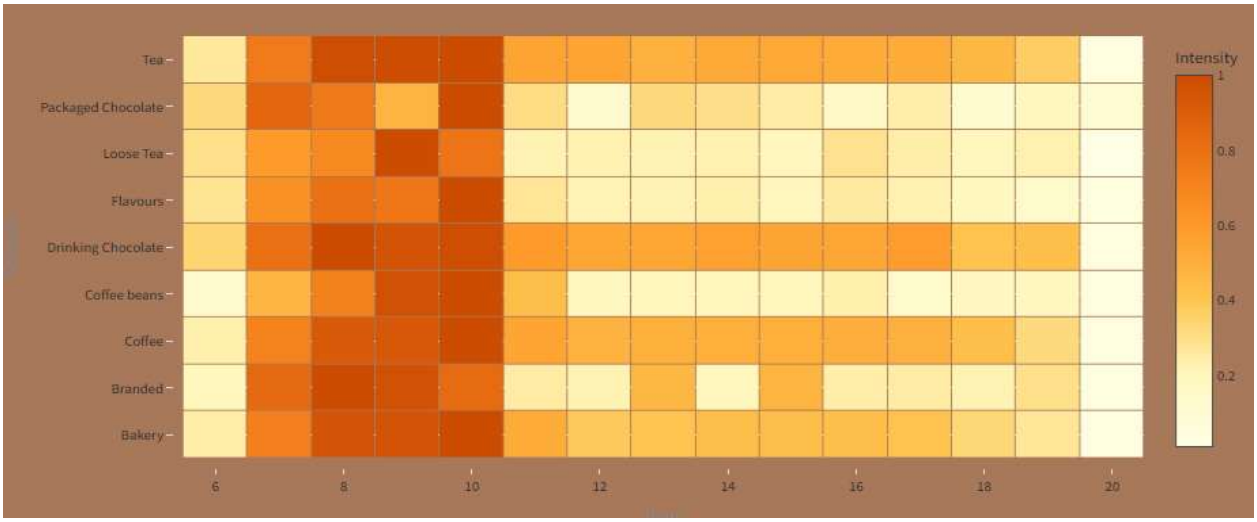

Figure 5: Heatmap showing revenue intensity across hours and categories

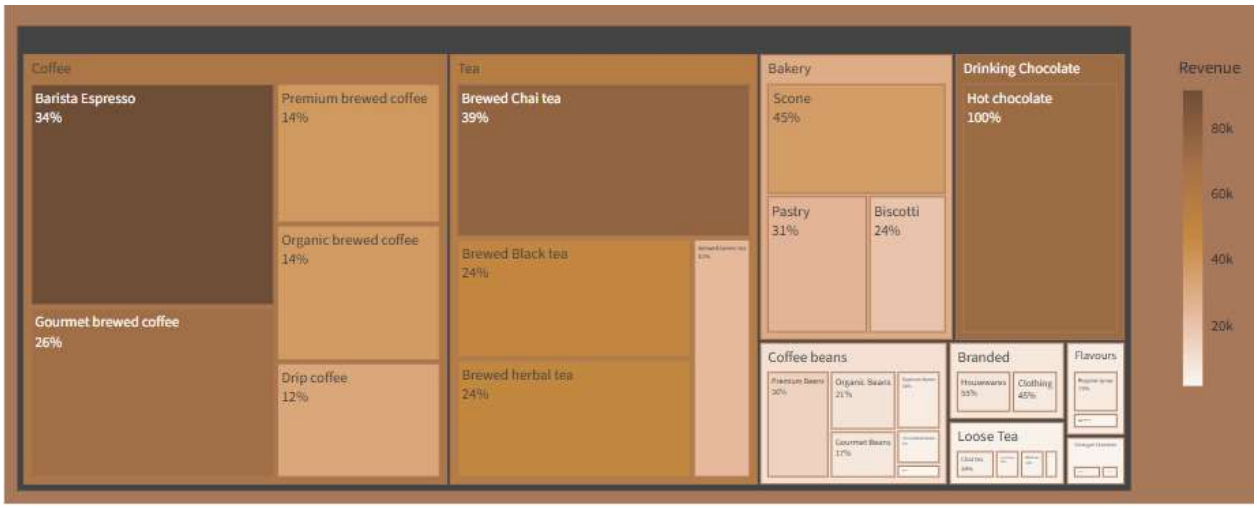

Figure 6: Revenue contribution across categories and product types

7. Recommendations

🚀 Strategic Actions:

- Focus marketing on top-performing products
- Reduce or eliminate low-impact SKUs
- Bundle slow-moving products
- Optimize pricing for high-volume items
- Improve inventory planning based on demand

8. Conclusion

This study demonstrates the importance of product-level analytics in retail decision-making. By identifying revenue concentration and product performance patterns, Afficionado Coffee Roasters can optimize its menu, enhance operational efficiency, and improve profitability. Data-driven strategies provide a competitive advantage in a highly dynamic retail environment.

9. Future Work

- Integrate real-time data analysis
- Apply machine learning for demand forecasting
- Enhance dashboard interactivity
- Expand analysis to customer segmentation